

Green-Lagrange Strain Tensor

right Cauchy-Green: $\underline{\underline{C}} = \underline{\underline{F}}^T \underline{\underline{F}}$

$\Rightarrow$ quantifies stretch & shear

$$\lambda(\hat{\underline{\underline{X}}}) = |\underline{\underline{dx}}| / |\underline{\underline{dX}}| = \sqrt{\hat{\underline{\underline{X}}} \cdot \underline{\underline{C}} \hat{\underline{\underline{X}}}}$$

No deformation: $\underline{\underline{x}} = \varphi(\underline{\underline{X}}) = \underline{\underline{X}} \Rightarrow \underline{\underline{F}} = \underline{\underline{C}} = \underline{\underline{I}} \Rightarrow \lambda = 1$

Constitutive law: $\underline{\underline{\sigma}} = \underline{\underline{\sigma}}(\underline{\underline{C}})$

no deformation = no stress

$\Rightarrow$ Strain tensor zero in absence of deformation

1D: engineering strain: $e = \lambda - 1$

3D Green-Lagrange strain tensor

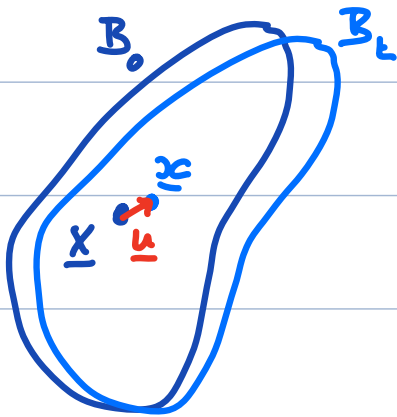
$$\underline{\underline{E}} = \frac{1}{2} (\underline{\underline{C}} - \underline{\underline{I}})$$

no deformation: $\underline{\underline{F}} = \underline{\underline{C}} = \underline{\underline{I}} \Rightarrow \underline{\underline{E}} = \underline{\underline{0}}$

relation to infinitesimal strain tensor

Small displacements

natural to use displacement



$$\underline{\bar{u}} = \underline{\bar{x}} - \underline{x}$$

$$\underline{\bar{x}} = \varphi(\underline{x})$$

$$\nabla \underline{\bar{u}} = \nabla(\varphi(\underline{x}) - \underline{x}) = \nabla \varphi - \underline{\underline{I}}$$

$$\underline{\underline{F}} = \underline{\underline{F}} - \underline{\underline{I}} \equiv \underline{\underline{H}}$$

Small deformation: $|\underline{\underline{\nabla u}}| = \sqrt{\underline{\underline{\nabla u}} : \underline{\underline{\nabla u}}} = \delta \ll 1$

Linearize Cauchy-Green

$$\underline{\underline{F}} = \underline{\underline{I}} + \underline{\underline{\nabla u}}$$

$$\begin{aligned} \underline{\underline{C}} &= \underline{\underline{F}}^T \underline{\underline{F}} = (\underline{\underline{I}} + \underline{\underline{\nabla u}})^T (\underline{\underline{I}} + \underline{\underline{\nabla u}}) \\ &= (\underline{\underline{I}} + \underline{\underline{\nabla u}}^T) (\underline{\underline{I}} + \underline{\underline{\nabla u}}) \\ &= \underline{\underline{I}} + \underline{\underline{\nabla u}} + \underline{\underline{\nabla u}}^T + \underbrace{\underline{\underline{\nabla u}}^T \underline{\underline{\nabla u}}}_{O(\delta^2)} \end{aligned}$$

$$\Rightarrow \underline{\underline{C}} \approx \underline{\underline{I}} + \underline{\underline{\nabla u}} + \underline{\underline{\nabla u}}^T$$

Linearize Euler-Lagrange

$$\underline{\underline{\mathbf{E}}} = \frac{1}{2} (\underline{\underline{\mathbf{C}}} - \underline{\underline{\mathbf{I}}}) = \frac{1}{2} (\underline{\underline{\mathbf{I}}} + \nabla \underline{\underline{\mathbf{u}}} + \nabla \underline{\underline{\mathbf{u}}}^T - \underline{\underline{\mathbf{I}}}) + O(\delta^2)$$

$$\underline{\underline{\mathbf{E}}} \approx \frac{1}{2} (\nabla \underline{\underline{\mathbf{u}}} + \nabla \underline{\underline{\mathbf{u}}}^T)$$

Infinitesimal strain tensor:

$$\underline{\underline{\mathbf{E}}} = \frac{1}{2} (\nabla \underline{\underline{\mathbf{u}}} + \nabla \underline{\underline{\mathbf{u}}}^T)$$

$$\underline{\underline{\mathbf{E}}} = \text{sym}(\nabla \underline{\underline{\mathbf{u}}})$$

Infinitesimal Stretch & Rotation

Linearize right stretch:

$$\underline{\underline{\mathbf{U}}} = \sqrt{\underline{\underline{\mathbf{C}}}} = \left(\underline{\underline{\mathbf{I}}} + \nabla \underline{\underline{\mathbf{u}}} + \nabla \underline{\underline{\mathbf{u}}}^T + \underbrace{\nabla \underline{\underline{\mathbf{u}}}^T \nabla \underline{\underline{\mathbf{u}}}}_{O(\delta^2)} \right)^{\frac{1}{2}}$$

Note: for $\underline{\underline{\mathbf{A}}} = \underline{\underline{\mathbf{A}}}^T \quad m \in \mathbb{R}$

$$(\underline{\underline{\mathbf{I}}} + \underline{\underline{\mathbf{A}}})^m = \underline{\underline{\mathbf{I}}} + m \underline{\underline{\mathbf{A}}} + O(|\underline{\underline{\mathbf{A}}}|^2)$$

shown with Taylor's expansion in principal frame

here $\underline{\underline{\mathbf{A}}} = \nabla \underline{\underline{\mathbf{u}}} + \nabla \underline{\underline{\mathbf{u}}}^T + O(\delta^2)$ and $m = \frac{1}{2}$

$$\Rightarrow \underline{\underline{\mathbf{U}}} = \underline{\underline{\mathbf{I}}} + \frac{1}{2} (\nabla \underline{\underline{\mathbf{u}}} + \nabla \underline{\underline{\mathbf{u}}}^T) + O(\delta^2) \approx \underline{\underline{\mathbf{I}}} + \underline{\underline{\mathbf{E}}}$$

Similarly $\underline{\underline{V}} = \sqrt{\underline{\underline{F}}\underline{\underline{F}}^T} = \underline{\underline{I}} + \frac{1}{2}(\nabla \underline{u} + \nabla \underline{u}^T) = \underline{\underline{I}} + \underline{\underline{\epsilon}}$

Note: $\underline{\underline{C}} \approx \underline{\underline{I}} + \underline{\underline{\epsilon}}$, $\underline{\underline{U}} \approx \underline{\underline{I}} + \underline{\underline{\epsilon}}$, $\underline{\underline{V}} \approx \underline{\underline{I}} + \underline{\underline{\epsilon}}$
 $\Rightarrow$ same linearization (Why?)

Linearize rotation

$$\begin{aligned}\underline{\underline{R}} &= \underline{\underline{F}} \underline{\underline{U}}^{-1} = (\underline{\underline{I}} + \nabla \underline{u})(\underline{\underline{I}} + \underline{\underline{\epsilon}})^{-1} & \underline{\underline{\epsilon}} &= O(\delta) \\ &= (\underline{\underline{I}} + \nabla \underline{u})(\underline{\underline{I}} - \underline{\underline{\epsilon}}) + O(\delta^2) \\ &= \underline{\underline{I}} - \frac{1}{2}(\nabla \underline{u} + \nabla \underline{u}^T) + \nabla \underline{u} + O(\delta^2) \\ &= \underline{\underline{I}} + \frac{1}{2}(\nabla \underline{u} - \nabla \underline{u}^T) + O(\delta^2) \\ \underline{\underline{R}} &\approx \underline{\underline{I}} + \frac{1}{2}(\nabla \underline{u} - \nabla \underline{u}^T) \\ &= \underline{\underline{I}} + \underline{\underline{\omega}}\end{aligned}$$

Infinitesimal Rotation Tensor: $\underline{\underline{\omega}} = \frac{1}{2}(\nabla \underline{u} - \nabla \underline{u}^T)$

$\underline{\underline{\omega}} = \text{skew}(\nabla \underline{u}) = \text{axial tensor}$

axis of rotation: $a_j = \frac{1}{2} \epsilon_{mjn} \omega_{mn}$

Infinitesimal rotation $\Rightarrow$ skew matrix!

Linearization of $\underline{\underline{F}} = \underline{\underline{R}} \underline{\underline{U}}$

$$\underline{\underline{R}} = \underline{\underline{I}} + \underline{\underline{\omega}} \quad \text{and} \quad \underline{\underline{U}} \approx \underline{\underline{I}} + \underline{\underline{\varepsilon}}$$

$$\underline{\underline{F}} = \underline{\underline{R}} \underline{\underline{U}} = (\underline{\underline{I}} + \underline{\underline{\varepsilon}})(\underline{\underline{I}} + \underline{\underline{\omega}}) = \underline{\underline{I}} + \underline{\underline{\varepsilon}} + \underline{\underline{\omega}} + O(|H^2|)$$

$$\Rightarrow \boxed{\underline{\underline{F}} = \underbrace{\underline{\underline{R}} \underline{\underline{U}}}_{\text{multiplicative}} \approx \underbrace{\underline{\underline{I}} + \underline{\underline{\varepsilon}} + \underline{\underline{\omega}}}_{\text{additive}}}$$

$$\underline{\underline{F}} = \underline{\underline{I}} + \nabla \underline{\underline{u}} = \underline{\underline{I}} + \text{sym}(\nabla \underline{\underline{u}}) + \text{skew}(\nabla \underline{\underline{u}})$$

$\Rightarrow$ nature of the rotation-stretch decomposition changes from multiplicative to additive! ∇

$\Rightarrow$ addition commutes but multiplication does not!

Finite theory: right and left version

Infinitesimal theory no difference.

Interpretation of components of $\underline{\underline{\epsilon}}$

Start from $\underline{\underline{C}}$

$$\underline{\underline{\epsilon}} = \underline{\underline{E}} = \frac{1}{2} (\underline{\underline{C}} - \underline{\underline{I}}) + O(\delta^2) \quad |\nabla \underline{u}| = \delta$$

$$\Rightarrow \underline{\underline{C}} \approx \underline{\underline{I}} + 2 \underline{\underline{\epsilon}}$$

I) Diagonal components

$$C_{ii} = 1 + 2 \epsilon_{ii}$$

$$\sqrt{C_{ii}} = \lambda(\underline{e}_i)$$

$$\lambda(\underline{e}_i) = \sqrt{1 + 2 \epsilon_{ii}}$$

Expand in Taylor-Series: $\sqrt{1-x} = 1 + \frac{x}{2} - \frac{x^2}{8} + \dots$

where $x = 2 \epsilon_{ii}$ s.t. $\lambda(\underline{e}_i) = 1 + \epsilon_{ii}$

$\Rightarrow$ $\epsilon_{ii} \approx \lambda(\underline{e}_i) - 1$ engineering strain
in coordinate directions

$$\lambda = \frac{|y-x|}{|Y-X|} = \frac{\ell}{L} \text{ stretch}$$

$$\lambda - 1 = \frac{|y-x| - |Y-X|}{|Y-X|} = \frac{\ell - L}{L} = \frac{\Delta \ell}{L}$$

$\epsilon_{ii} =$ relative change in length

II) Off-Diagonal Components

$$C_{ij} = \lambda(\underline{e}_i) \lambda(\underline{e}_j) \sin(\gamma_{ij}) \quad \gamma_{ij} = \gamma(\underline{e}_i, \underline{e}_j)$$

$$C_{ij} \approx 2 \varepsilon_{ij} \quad i \neq j$$

$$\varepsilon_{ij} \approx \frac{1}{2} \lambda(\underline{e}_i) \lambda(\underline{e}_j) \sin(\gamma_{ij})$$

3 terms:

$$\lambda(\underline{e}_i) = 1 + \varepsilon_{ii} = 1 + O(\epsilon)$$

$$\lambda(\underline{e}_j) = 1 + \varepsilon_{jj} = 1 + O(\epsilon)$$

$$\sin(\gamma_{ij}) = O(\epsilon)$$

$$\begin{aligned} \varepsilon_{ij} &= \frac{1}{2} (1 + \varepsilon_{ii}) (1 + \varepsilon_{jj}) \sin(\gamma_{ij}) \\ &= \frac{1}{2} \left[\sin(\gamma_{ij}) + \underbrace{\varepsilon_{ii} \sin(\gamma_{ij})}_{O(\epsilon^2)} + \underbrace{\varepsilon_{jj} \sin(\gamma_{ij})}_{O(\epsilon^2)} + \underbrace{\varepsilon_{ii} \varepsilon_{jj} \sin(\gamma_{ij})}_{O(\epsilon^3)} \right] \end{aligned}$$

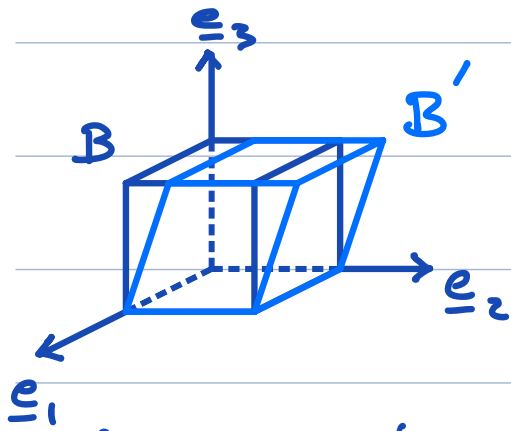
$$\Rightarrow \varepsilon_{ij} = \frac{1}{2} \sin(\gamma(\underline{e}_i, \underline{e}_j))$$

Infinitesimal strain $\Rightarrow \gamma \ll 1 \Rightarrow \sin \gamma \approx \gamma$

$$\varepsilon_{ij} \approx \frac{1}{2} \gamma(\underline{e}_i, \underline{e}_j)$$

$\Rightarrow$ half shear angle between coord. directions

Example: Simple shear



$$\underline{x} = \underline{\varphi}(\underline{X}) = \begin{bmatrix} \varphi_1 \\ \varphi_2 \\ \varphi_3 \end{bmatrix} = \begin{bmatrix} X_1 \\ X_2 + \alpha X_3 \\ X_3 \end{bmatrix} \quad \alpha > 0$$

large def:
(last lecture)

$$\underline{C} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & \alpha \\ 0 & \alpha & 1 + \alpha^2 \end{bmatrix}$$

Infinitesimal:

$$\underline{u} = \underline{x} - \underline{X} = \begin{bmatrix} 0 \\ \alpha X_3 \\ 0 \end{bmatrix}$$

$$\nabla \underline{u} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \alpha \\ 0 & 0 & 0 \end{bmatrix}$$

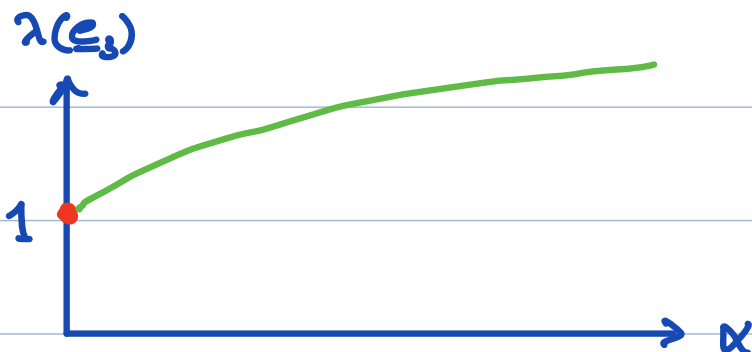
$$\underline{\underline{\epsilon}} = \frac{1}{2} (\nabla \underline{u} + \nabla \underline{u}^T) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & \alpha/2 \\ 0 & \alpha/2 & 0 \end{bmatrix}$$

Infinitesimal stretches: $\epsilon_{ii} = \lambda(\underline{e}_i) - 1 = 0$

$\lambda(\underline{e}_i) = 1$ no stretch in any coord. direction!

Finite stretches: $C_{ii} = \lambda^2(\underline{e}_i)$ $C_{33} = \lambda^2(\underline{e}_3) = 1 + \alpha^2$

$$\lambda(\underline{e}_3) = \sqrt{1 + \alpha^2}$$



- finite strain

- infinitesimal strain

Infinitesimal shear: $\varepsilon_{ij} \approx \frac{1}{2} \gamma(e_i, e_j)$
 $\gamma(e_2, e_3) = 2 \varepsilon_{23} = \alpha$

Finite shear: $\gamma(e_2, e_3) = a \sin\left(\frac{\alpha}{\sqrt{1+\alpha^2}}\right)$

